4. Business Requirements Document (BRD)

4.1 Project Overview

Project Name: Retail Returns & Reverse Logistics Optimization

Industry: Retail / E-Commerce

Project Role: Business Analyst

Tools: Excel, SQL, Power BI and Generative AI

Project Purpose

The purpose of this project is to analyze retail transaction and cancellation data to identify important patterns, understand their business impact, and recommend improvements to the returns and reverse-logistics process.

4.2 Current Business Situation

The retailer processes a large volume of online transactions. Some transactions are cancelled, creating additional operational and financial challenges.

Currently, the business does not have a centralized analytical view to easily understand:

- Which products experience higher cancellation activity
- Which customers have higher cancellation activity
- How cancellation activity changes over time
- Which transactions have higher financial impact
- Where improvements to the existing process may be required

4.3 Business Problem

The lack of structured analysis makes it difficult for management to identify the major areas contributing to cancellations and prioritize improvement opportunities.

The business therefore requires a data-driven analysis and reporting solution that can provide visibility into transaction and cancellation patterns.

4.4 Project Objectives

Objective 1

Analyze retail transaction data to identify cancellation patterns.

Objective 2

Identify products, customers, time periods, and transaction values associated with higher cancellation activity.

Objective 3

Develop a Power BI dashboard to provide management with clear visibility into important cancellation KPIs and trends.

Objective 4

Use AI-assisted analysis to identify meaningful patterns and support data-driven business recommendations.

4.5 Scope

In Scope

- Retail transaction data analysis
- Data cleaning and preparation using Excel
- SQL-based analysis
- Cancellation pattern analysis
- Product and customer analysis
- Time-based trend analysis
- Financial impact analysis
- Power BI dashboard
- AI-assisted insight generation
- Business recommendations
- Process improvement recommendations
- User stories and acceptance criteria

Out of Scope

- Development of a production-level returns management system
- Actual integration with an e-commerce platform
- Development of a machine-learning model
- Live customer transactions
- Actual implementation of recommended process changes

4.6 Functional Requirements

ID	Requirement
FR01	The solution shall allow analysis of transactions by product and product category.
FR02	The solution shall identify cancelled transactions.
FR03	The solution shall calculate cancellation rates and transaction counts.
FR04	The solution shall allow analysis of cancellation trends over time.
FR05	The solution shall identify customers with relatively high cancellation activity.
FR06	The solution shall calculate the financial value associated with cancelled transactions.
FR07	The Power BI dashboard shall provide filters for relevant business dimensions.
FR08	The solution shall provide insights and recommendations based on the identified patterns.

4.7 Non-Functional Requirements

ID	Requirement
NFR01	The dashboard should be easy for business users to understand.
NFR02	Calculations should be accurate and based on validated transaction data.
NFR03	The dashboard should provide clear and consistent KPI definitions.
NFR04	The analysis should be structured so that it can be updated when new data becomes available.
NFR05	The solution should present information in a clear and user-friendly format.

4.8 Key KPIs / Success Metrics

The following KPIs will be analyzed:

- Total Transactions
- Total Cancelled Transactions
- Cancellation Rate
- Total Transaction Value

- Cancelled Transaction Value
- Cancellation Rate by Product
- Cancellation Rate by Customer
- Cancellation Trend Over Time
- Average Transaction Value
- Top Products by Cancellation Value

Note: Actual KPI values will be calculated during the SQL and Power BI analysis using the UCI Online Retail II dataset. We will not assume or invent these values before analyzing the data.

4.9 Expected Outcome

The project is expected to provide the retailer with a clear understanding of cancellation patterns and their potential business impact.

The analysis will be used to identify priority areas and develop practical recommendations for improving the cancellation and reverse-logistics process.

4.10 Stakeholders & Roles

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Stakeholder	Role / Responsibility
Business Management	Review cancellation performance and make business decisions based on insights.
Operations Team	Investigate operational factors contributing to cancellations and implement process improvements.
E-commerce / Sales Team	Monitor transaction and product-level cancellation patterns.
Inventory / Fulfilment Team	Investigate product availability and fulfilment-related issues associated with cancellations.
Business Analyst	Gather requirements, analyze data, identify gaps, define KPIs, document requirements, and recommend improvements.
Data / BI Team	Maintain data preparation, SQL analysis, and Power BI reporting.
IT / System Team	Support future implementation or integration of proposed process and reporting changes.

4.11 Stakeholder Information Needs

- Management requires high-level cancellation KPIs and trends.
- Operations requires product, country, and time-based cancellation patterns.
- Inventory/Fulfilment teams require information that can help identify operational problem areas.
- Business Analysts require validated data and business requirements to identify gaps and recommend improvements.
- BI/Data teams require clearly defined metrics and data requirements for reporting.

4.12 Business Rules

ID	Business Rule
BR01	A transaction shall be considered cancelled when the Cancellation_Flag indicates a cancellation.
BR02	Cancellation Rate shall be calculated as Cancelled Transactions divided by Total Transactions, multiplied by 100.
BR03	Cancellation Value shall represent the transaction value associated with cancelled transactions.
BR04	Negative-quantity transactions shall be analyzed separately and shall not automatically be treated as cancellations unless supported by the cancellation flag.
BR05	Transactions with missing Customer IDs shall remain available for product, country, and transaction-level analysis but shall be excluded from customer-specific analysis.
BR06	KPI calculations shall use validated transaction data from the analyzed retail dataset.
BR07	AI-generated insights shall be treated as decision-support outputs and validated against the underlying Excel, SQL, and Power BI analysis before business decisions are made.
BR08	Business recommendations shall be based on identified data patterns and shall not be treated as confirmed root causes without further operational validation.

4.13 Data Requirements

ID	Data Requirement
DR01	The solution shall use retail transaction data containing invoice, product, quantity, date, price, customer, country, and cancellation information.
DR02	Each transaction should have a unique invoice/transaction reference where available.
DR03	Product information shall include StockCode and Description.
DR04	Transaction data shall include Quantity and Price for transaction-level analysis.
DR05	Customer ID shall be used for customer-level analysis where available.
DR06	Country shall be available for geographic cancellation analysis.
DR07	Invoice Date and Month shall support time-based analysis.
DR08	Cancellation Flag shall be used to identify cancelled transactions.
DR09	Transaction value shall be available to evaluate the financial impact of cancellations.
DR10	Missing, invalid, or inconsistent data shall be identified during data preparation and considered when interpreting results.
DR11	Data used for KPI reporting shall be validated through Excel and SQL analysis before being presented in Power BI.
DR12	AI analysis shall use validated analytical outputs rather than unverified raw assumptions.

4.14 Assumptions & Constraints

Assumptions

- The available transaction data is representative of the business activity being analyzed.
- Cancellation records identified through the available data are suitable for cancellation analysis.
- The business will validate identified patterns before implementing process changes.

- Power BI will be used as the primary reporting and visualization layer.
- AI-generated insights will be reviewed by a Business Analyst or business stakeholder before being used for decision-making.

Constraints

- The project uses historical transaction data rather than live transaction data.
- The dataset contains missing and inconsistent values that may affect some analyses.
- The project does not include development of a production-level returns management system.
- No live e-commerce or inventory-system integration is implemented.
- The proposed process improvements are recommendations and are not actually implemented within this project.
- AI is used for insight generation and decision support rather than autonomous decision-making.

This is consistent with the **Out of Scope** section already present in your BRD, particularly the exclusions around production-system development, live transactions, and actual implementation.

4.15 Gap Analysis

Area	**Current State (AS-IS)**	**Gap Identified**	**Proposed Improvement (TO-BE)**
Cancellation Monitoring	Cancellation information is available in transaction data but requires manual analysis.	No centralized cancellation monitoring.	Power BI dashboard with cancellation KPIs and trends.
Product Analysis	Product-level cancellation patterns are difficult to identify quickly.	Limited visibility into high-cancellation products.	Monitor product-level cancellation counts and rates.
Geographic Analysis	Country information exists but is not presented as a centralized business view.	Limited visibility into geographic patterns.	Provide country-level cancellation analysis.

Area	Current State (AS-IS)	Gap Identified	Proposed Improvement (TO-BE)
Time-based Analysis	Historical transactions require separate analysis to identify trends.	Cancellation trends are not easily monitored.	Provide monthly and yearly cancellation trend reporting.
Financial Impact	Transaction values exist within the dataset.	Cancellation volume alone may not show financial impact.	Track cancellation transaction value alongside cancellation counts.
Data Quality	Missing and inconsistent values exist in the source data.	Data-quality issues can affect analysis.	Perform data validation and document data-quality limitations.
Decision Support	Analysis requires manual interpretation of results.	Business users may spend additional time interpreting data.	Use AI-assisted insight generation based on validated analytical results.
Process Improvement	The existing data does not provide a structured improvement workflow.	Root causes and improvement opportunities require further investigation.	Define proposed process improvements and business requirements.

Gap Analysis Summary

The primary gap is the absence of a centralized, data-driven approach for monitoring cancellation performance and translating analytical findings into actionable business improvements. The proposed solution addresses this gap through structured data analysis, Power BI reporting, AI-assisted insights, and process improvement recommendations.

This extends the existing BRD's stated problem that management lacks a centralized analytical view of cancellation patterns and financial impact.

4.16 AS-IS Process

Current Process

The current process represents how cancellation information is handled before implementing the proposed analytical and process-improvement solution.

1. Customer places an online order.
2. Transaction information is recorded in the retail transaction system.
3. If an order is cancelled, the cancellation is recorded along with the transaction information.
4. Transaction data accumulates across products, customers, countries, dates, quantities, and transaction values.
5. Business users analyze the available data using manual or separate analytical methods.
6. Cancellation patterns are identified after analysis rather than through a centralized monitoring mechanism.
7. Management reviews the findings and identifies potential areas requiring investigation.
8. Any corrective action requires further investigation by the relevant operational teams.

AS-IS Process Flow

Customer Order

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Transaction Recorded

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Order Cancelled?

↙ ↘

No Yes

↓ ↓

Normal Cancellation

Processing Recorded

↘ ↙

Transaction Data

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Manual / Separate

Analysis

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Identify Patterns

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Management Review

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Further Investigation

AS-IS Process Gap

The current process does not provide a centralized mechanism to continuously monitor cancellation KPIs, identify high-risk areas, and translate analytical findings into structured improvement actions.

4.17 TO-BE Process

Proposed Future-State Process

The proposed process introduces centralized monitoring, structured analysis, and AI-assisted decision support.

1. Customer places an online order.
2. Transaction information is captured in the retail transaction system.
3. If an order is cancelled, the cancellation is recorded.
4. Transaction data is collected and validated.
5. SQL and Excel analysis are used to calculate cancellation KPIs and identify patterns.
6. Power BI provides a centralized dashboard for monitoring cancellation performance.
7. Business users use filters to investigate products, countries, and time periods.
8. AI-assisted analysis converts validated analytical results into business insights.

9. Business stakeholders review the identified patterns and potential causes.
10. Relevant operational teams investigate root causes.
11. Corrective actions are prioritized based on business impact.
12. Cancellation KPIs are monitored to evaluate whether performance improves.

TO-BE Process Flow

Customer Order

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Transaction Recorded

↓

Order Cancelled?

↙ ↘

No Yes

↓ ↓

Normal Cancellation

Processing Recorded

↘ ↙

Validated Data

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Excel / SQL

Analysis

↓

Power BI Dashboard

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Identify High-Risk

Products / Markets

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AI-Assisted Insights

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Business Validation

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Root Cause Investigation

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Prioritize Improvements

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Implement Actions

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Monitor Cancellation KPIs

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Continuous Improvement

Expected Improvement

The TO-BE process provides a structured path from transaction data to business action. It improves visibility, supports faster identification of problem areas, and enables management to monitor whether recommended improvements are reducing cancellation activity.

This directly extends your existing project objective of using Power BI and AI-assisted analysis to provide visibility and support data-driven recommendations.

4.18 User Requirements & User Stories

User Requirements

ID	User Requirement
UR01	Business users should be able to view overall cancellation KPIs.
UR02	Business users should be able to analyze cancellation trends over time.
UR03	Business users should be able to identify products with high cancellation activity.
UR04	Business users should be able to analyze cancellation activity by country.

ID	User Requirement
UR05	Business users should be able to evaluate the financial impact of cancellations.
UR06	Business users should be able to filter dashboard information by relevant dimensions.
UR07	Business stakeholders should be able to use validated analytical results to support operational decisions.
UR08	Business users should be able to review AI-generated insights based on validated analysis.

User Stories

US01 — View Cancellation KPIs

As a Business Manager, I want to view total transactions, cancelled transactions, cancellation rate, and cancellation value so that I can understand overall cancellation performance.

Acceptance Criteria:

- Total transactions are displayed.
- Cancelled transactions are displayed.
- Cancellation rate is displayed as a percentage.
- Cancellation value is displayed.
- KPI values are based on validated data.

US02 — Analyze Cancellation Trends

As an Operations Manager, I want to view cancellation trends over time so that I can identify periods with increased cancellation activity.

Acceptance Criteria:

- Monthly cancellation trends are displayed.
- Users can identify periods with higher cancellation activity.
- Data is presented using a clear visual trend.

US03 — Identify High-Cancellation Products

As an Inventory/Operations Manager, I want to identify products with high cancellation activity so that I can investigate potential operational issues.

Acceptance Criteria:

- Products are ranked by cancellation activity.
- Top cancellation products can be identified.
- The analysis is based on validated transaction data.

US04 — Analyze Geographic Patterns

As a Business Manager, I want to analyze cancellations by country so that I can identify markets requiring further investigation.

Acceptance Criteria:

- Cancellation activity can be viewed by country.
- Countries can be compared based on cancellation volume.
- Relevant country filters are available.

US05 — Analyze Financial Impact

As a Finance/Business Manager, I want to understand the transaction value associated with cancellations so that I can prioritize high-impact areas.

Acceptance Criteria:

- Cancellation value is calculated.
- High-value cancellation areas can be identified.
- The calculation uses validated transaction-value data.

US06 — Receive AI-Assisted Insights

As a Business Analyst, I want AI to summarize validated analytical findings so that stakeholders can understand important patterns and potential actions more quickly.

Acceptance Criteria:

- AI receives validated analytical outputs.
- AI identifies relevant patterns.
- AI clearly distinguishes observations from possible hypotheses.
- AI-generated recommendations are reviewed by a human before implementation.

US07 — Monitor Improvement

As a Business Manager, I want to monitor cancellation KPIs over time so that I can evaluate whether corrective actions are improving performance.

Acceptance Criteria:

- Cancellation rate can be monitored over time.
- Historical performance can be compared.
- Results can be used to support continuous improvement.

This completes the **User Requirements & User Stories** section. Your existing BRD already lists user stories and acceptance criteria as part of the project's scope, so this section fills that gap rather than changing your original BRD direction.

4.19 Acceptance Criteria

ID	Requirement / Feature	Acceptance Criteria
AC01	KPI Dashboard	Dashboard displays Total Transactions, Cancelled Transactions, Cancellation Rate, and Cancellation Value.
AC02	Cancellation Trend	User can view cancellation activity across different months.
AC03	Product Analysis	User can identify products with high cancellation activity.
AC04	Country Analysis	User can compare cancellation activity across countries.
AC05	Year Comparison	User can compare cancellation rates between 2009–2010 and 2010–2011.
AC06	Dashboard Filtering	User can filter dashboard results by Year Period and Country.

ID	Requirement / Feature	Acceptance Criteria
AC07	Financial Analysis	Cancellation transaction value is calculated and displayed correctly.
AC08	Data Validation	Dashboard calculations are based on validated Excel and SQL analysis.
AC09	AI Insights	AI-generated insights are based on validated analytical outputs and do not replace the underlying calculations.
AC10	Business Recommendations	Recommendations are linked to identified analytical findings and are subject to stakeholder validation.

4.20 Requirements Traceability Matrix

Requirement ID	Business Objective	Solution Component	Validation
FR01	Identify cancellation patterns	SQL + Power BI	Product analysis
FR02	Identify cancelled transactions	SQL + Power BI	Cancellation KPI
FR03	Measure cancellation performance	SQL + Power BI	Cancellation rate
FR04	Analyze trends over time	Power BI	Monthly trend
FR05	Identify high-cancellation areas	SQL + Power BI	Product/Country analysis
FR06	Measure financial impact	SQL + Power BI	Cancellation value
FR07	Enable business filtering	Power BI	Year and Country slicers
FR08	Support business decisions	AI + BA analysis	AI insights and recommendations
UR01	Monitor KPIs	Power BI	KPI cards

Requirement ID	Business Objective	Solution Component	Validation
UR02	Identify trends	Power BI	Monthly cancellation chart
UR03	Identify high-risk products	SQL + Power BI	Top 10 products
UR04	Identify geographic patterns	SQL + Power BI	Country analysis
UR05	Evaluate financial impact	SQL + Power BI	Cancellation value
UR06	Filter information	Power BI	Dashboard slicers
UR07	Support decisions	SQL + Power BI	Validated findings
UR08	Generate insights	Generative AI	AI insights document

4.21 Requirements Summary

The requirements defined in this BRD establish a complete analytical and decision-support solution for retail cancellation management. The solution connects business objectives with data analysis, reporting, AI-assisted insights, and proposed process improvements.

The requirements are designed to support the project's overall objective of improving visibility into cancellation patterns and helping stakeholders identify priority areas for operational improvement.